CNeuro

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1 Introduction

- Case 1 .csv yielded 2 clean RP epochs from 3 events, 1 was rejected due to artifacts.
- Case 3 .csv yielded 3 clean RP epochs from 7 events (57% rejection) and only 1 clean CNV epoch from 2 warning action pairs too few for reliable averaging.
- Case 2 showed that even small movements like shaking your head or smiling create huge spikes in the EEG signal that completely hide the brain activity we actually care about.
- When we looked at the frequency content of the signals, Cases 1 and 3 looked normal and healthy, while Case 2 was full of low-frequency noise caused by all the movements.
- Overall, epoch counts are too low across all cases, more trials per participant are needed in the next recording session.

2 Data Collection

2.1 Equipment

EEG was recorded using the **Unicorn Hybrid Black** wireless headset

2.2 Participants and Recordings

Three separate recording sessions were conducted on **November 2025**, each producing a raw CSV file from the Unicorn recorder:

Case	Purpose	Duration (min)
Case 1	RP analysis – cued and spontaneous braking events	8.08
Case 2	Artifact demonstration – deliberate motion/facial noise	11.15
Case 3	Both RP and CNV – instruction-cued and free braking	8.38

All events were marked by the manual writing of time in (mm:ss) time that each driving event took place within the file.

3 Signal Processing

3.1 Preprocessing

After loading each CSV, two filters were applied sequentially to every EEG channel:

1. **50 Hz notch filter** (IIR, $Q = 30$)
2. **Bandpass filter** (Butterworth 4th order, 0.1–10 Hz)

3.2 Epoching

The epochs were clipped from the continuous filtered signal, centered on each annotated event. Two epoch windows were defined:

- **RP window:** -2.0 s to $+3.0$ s around the motor onset (1251 samples at 250 Hz).
- **CNV window:** -3.0 s to $+2.0$ s around the action onset (1251 samples).

3.3 Baseline Correction

-0.5 to 0.0 s for RP; -2.5 to -2.0 s for CNV.

3.4 Artifact Rejection

1. **EEG amplitude:** epochs where peak-to-peak amplitude on any channel exceeded **120** μ**V**
2. **Motion RMS:** Epochs with a robust Z-score exceeding 4.5 were discarded.

4 Results and Figures

4.1 Raw EEG with Event Markers

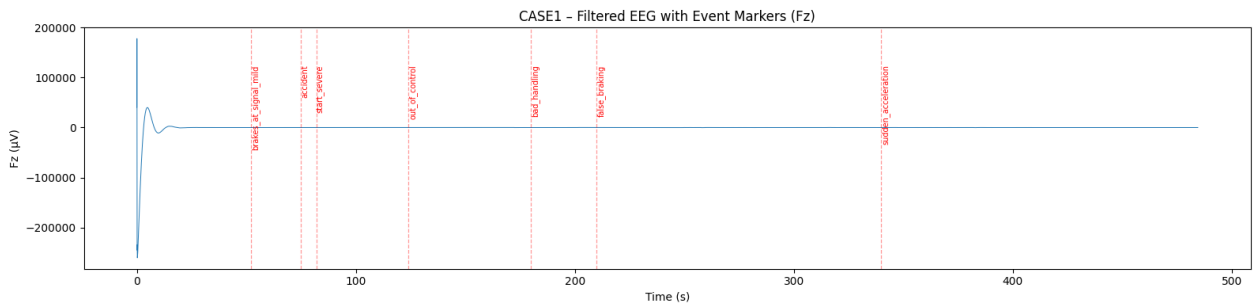


Figure 1: Case 1: Filtered EEG (Fz, 0.1–10 Hz) across the full 8.08-minute recording. Red dashed lines mark the 7 annotated events. at 124 s.

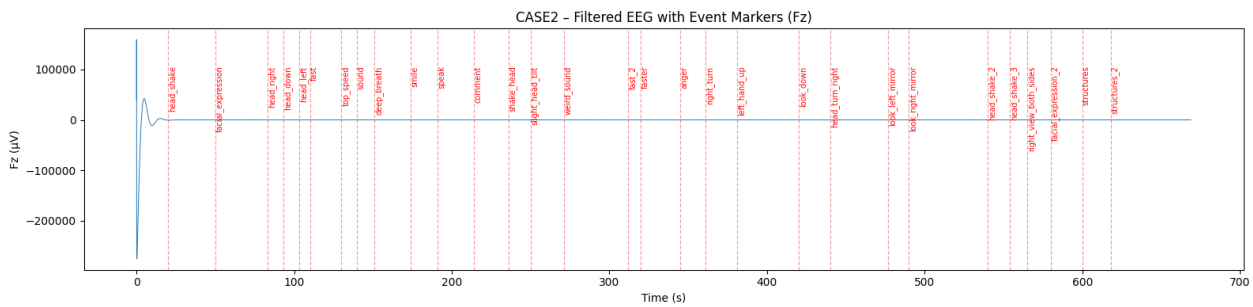


Figure 2: Case 2: Filtered EEG (Fz) across 11.15 minutes.

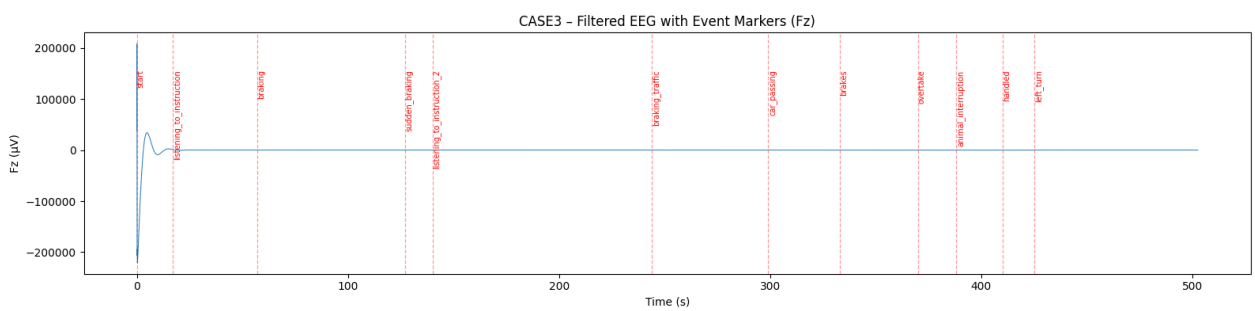


Figure 3: Case 3: Filtered EEG (Fz) across 8.38 minutes with 12 events marked.

4.2 Readiness Potential – Case 1

Three RP events were defined: `brakes_at_signal_mild`, `false_braking`, and `sudden_acceleration`. After artifact rejection **2 out of 3 epochs** were retained.

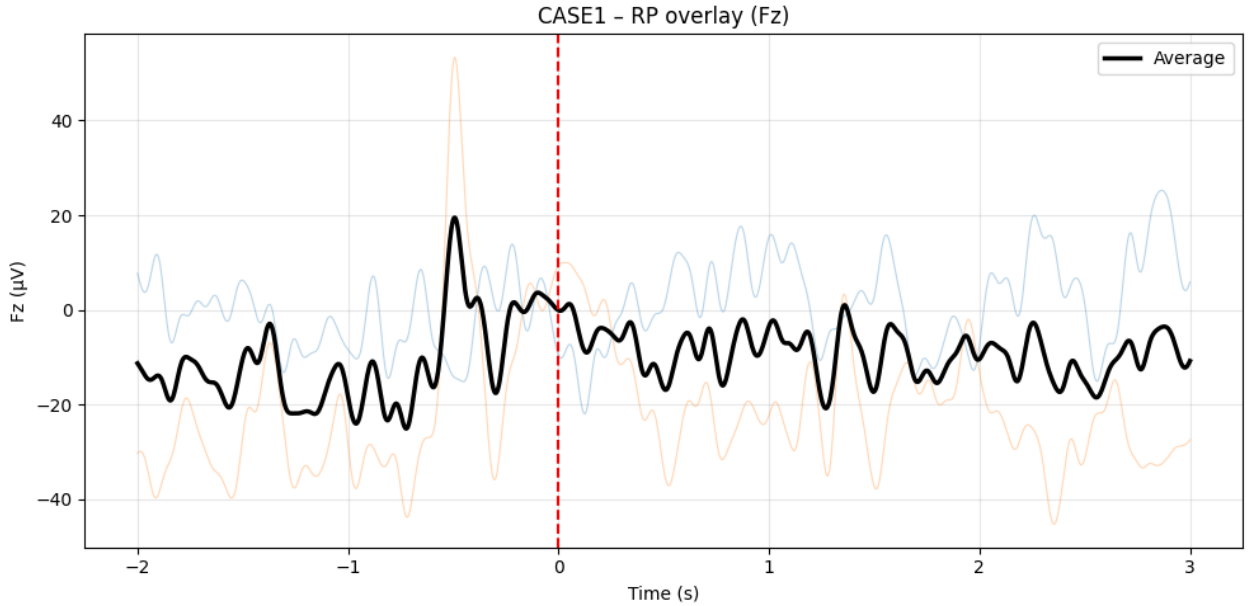


Figure 4: Case 1 RP: epoch overlay at Fz (2 clean epochs).

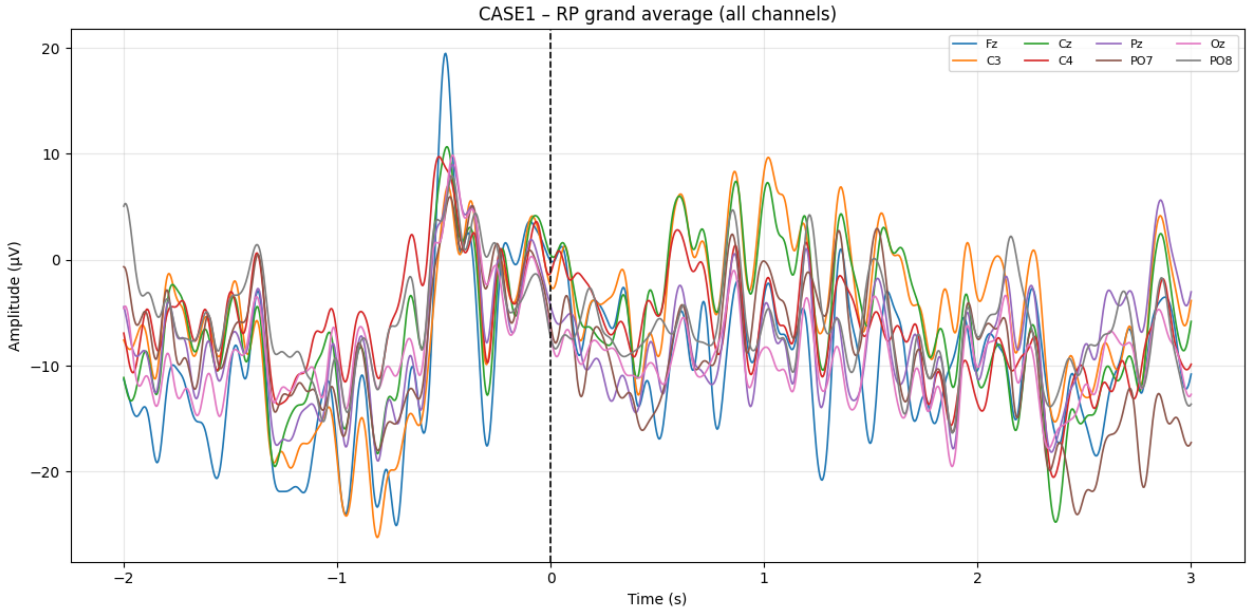


Figure 5: Case 1 RP: grand average across all 8 channels (2 epochs).

4.3 Readiness Potential – Case 3

Seven RP‘s were identified (braking, sudden braking, traffic braking, brakes, overtake, animal, and left turn). Then the request is rejected. We kept 3 out of the 7 epochs. The rejection rate of 57% represents a naturalistic driving environment when participants moved their heads and a free circulation of the bodies.

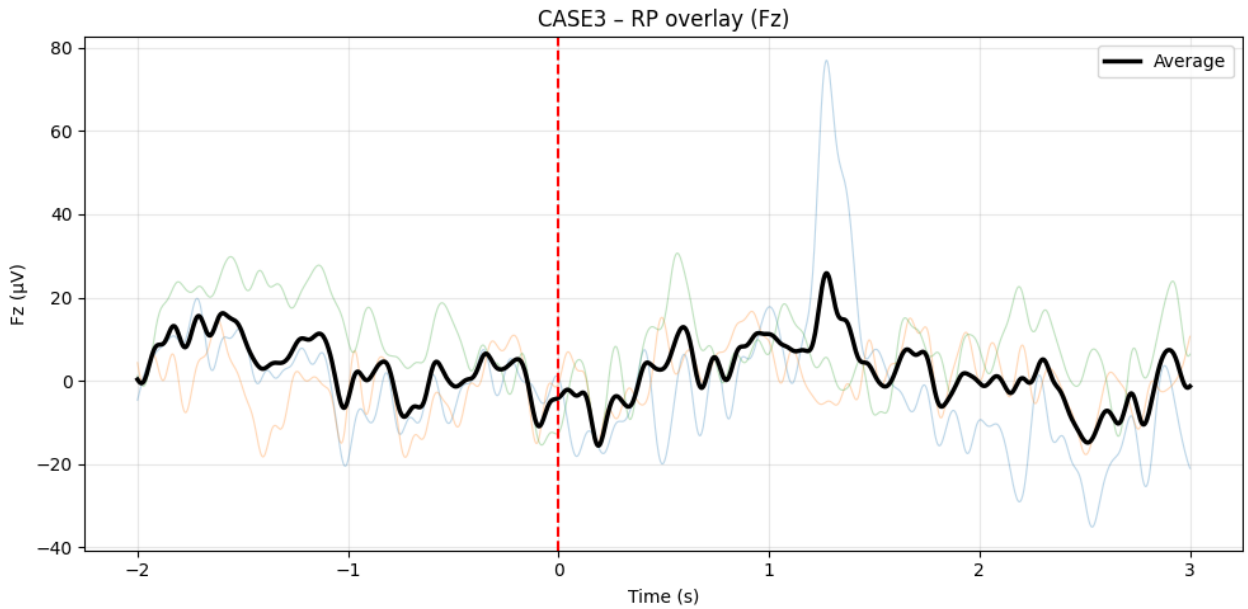


Figure 6

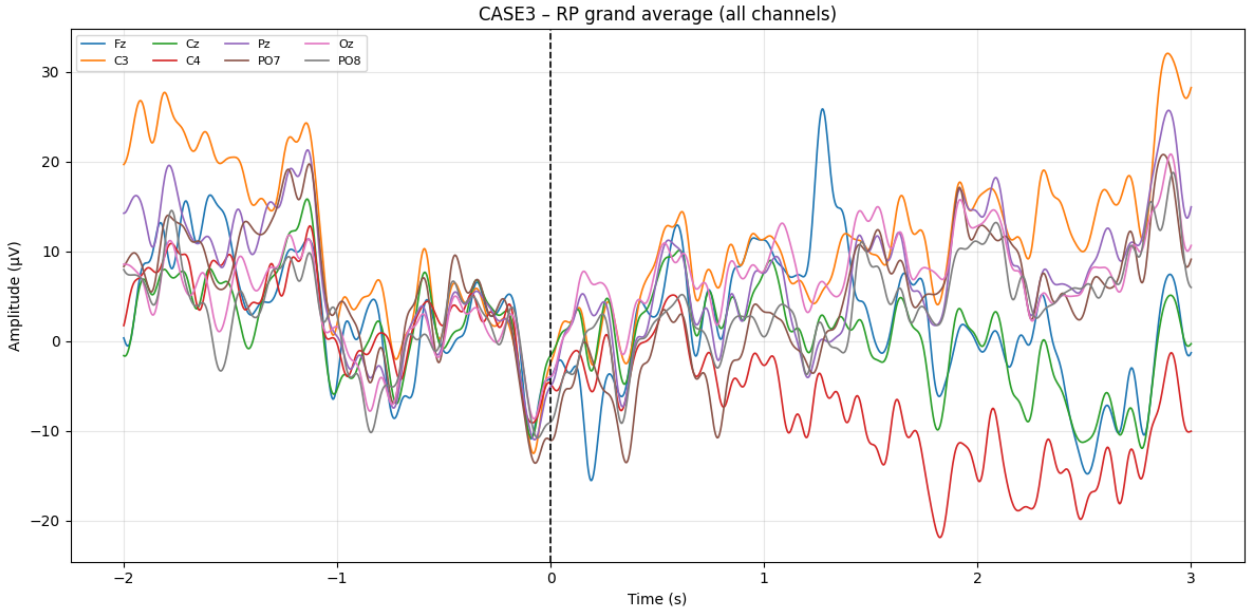

CASE3 – RP grand average (all channels)
Fz
C3
Cz
C4
Pz
PO7
Oz
PO8
Amplitude (µV)
Time (s)
30
20
10
0
−10
−20
−2
−1
0
1
2
3

Figure 7

4.4 Contingent Negative Variation – Case 3

Two warning–action pairs were used:

- `listening_to_instruction` → `braking`
- `listening_to_instruction_2` → `braking_traffic`

After rejection **1 out of 2 epochs** was retained.

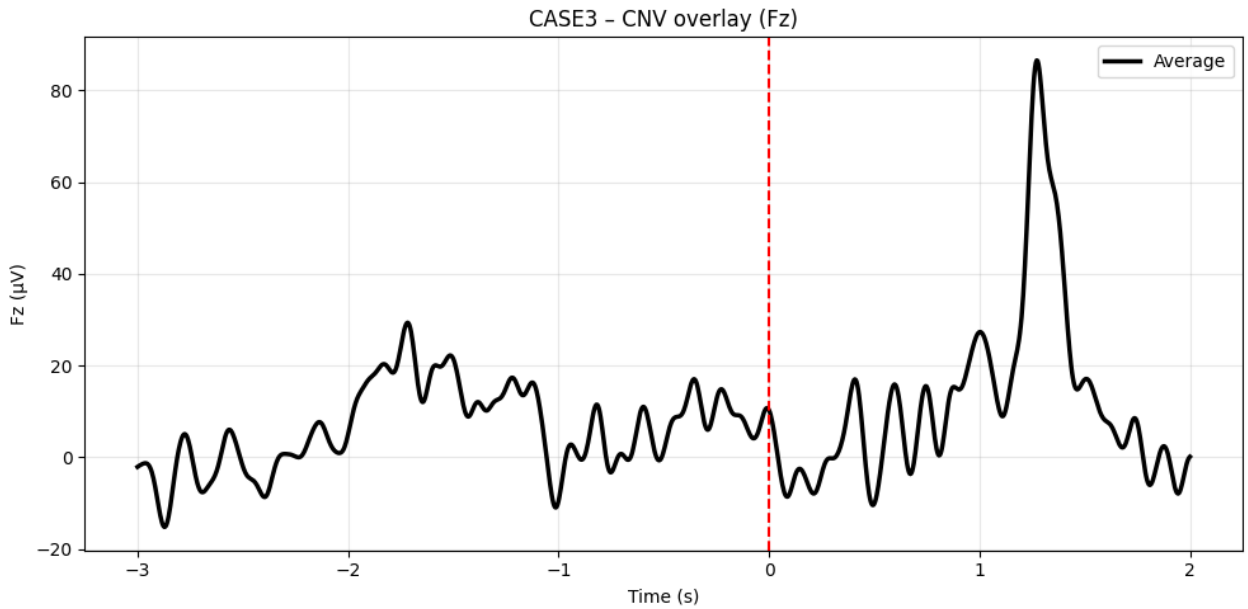


Figure 8

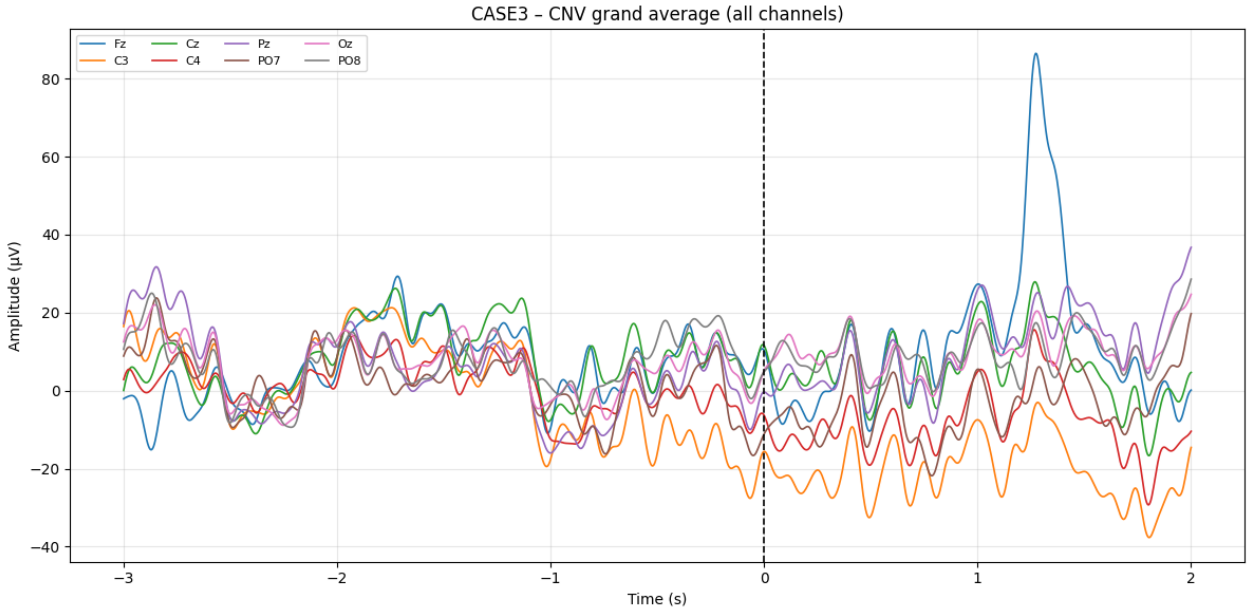

CASE3 – CNV grand average (all channels)
Fz
C3
Cz
C4
Pz
PO7
Oz
PO8
Amplitude (µV)
Time (s)
80
60
40
20
0
−20
−40
−3
−2
−1
0
1
2

Figure 9

4.5 Case 2 – Artifact Demonstration

Case 2 documents common EEG contamination sources. Eleven epochs were extracted around deliberate head movements, facial expressions, and vocalisations (`head_shake`, `head_right/down/left`, `deep_breath`, `smile`, `speak`, `comment`, `shake_head`, `slight_head_tilt`, `left_hand_up`).

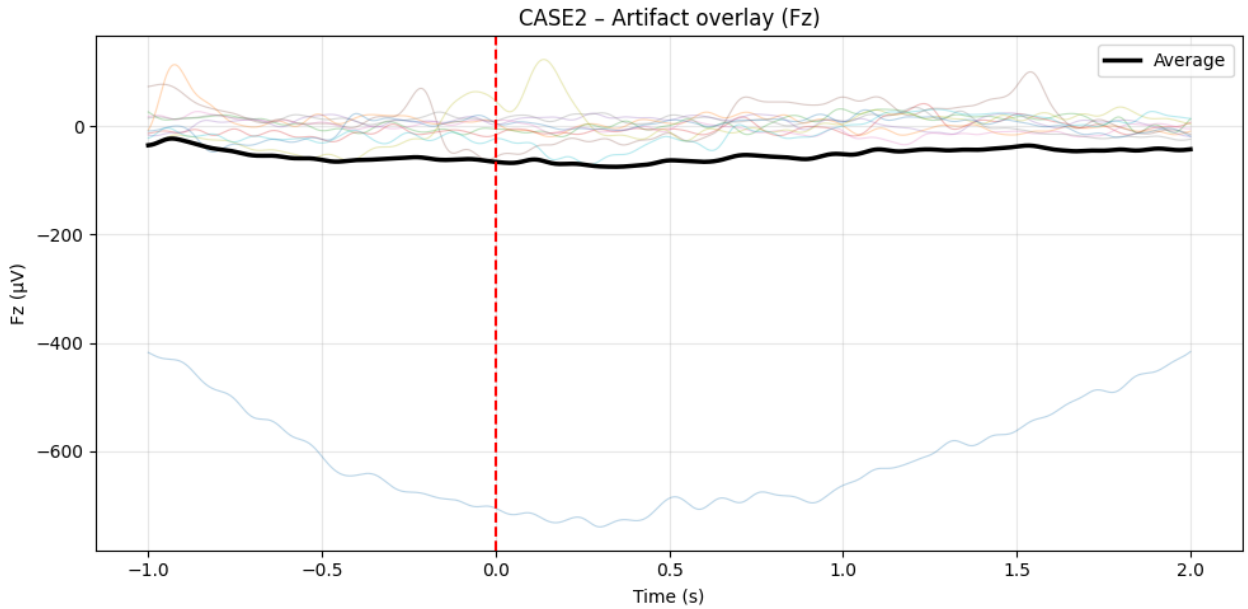


Figure 10: .

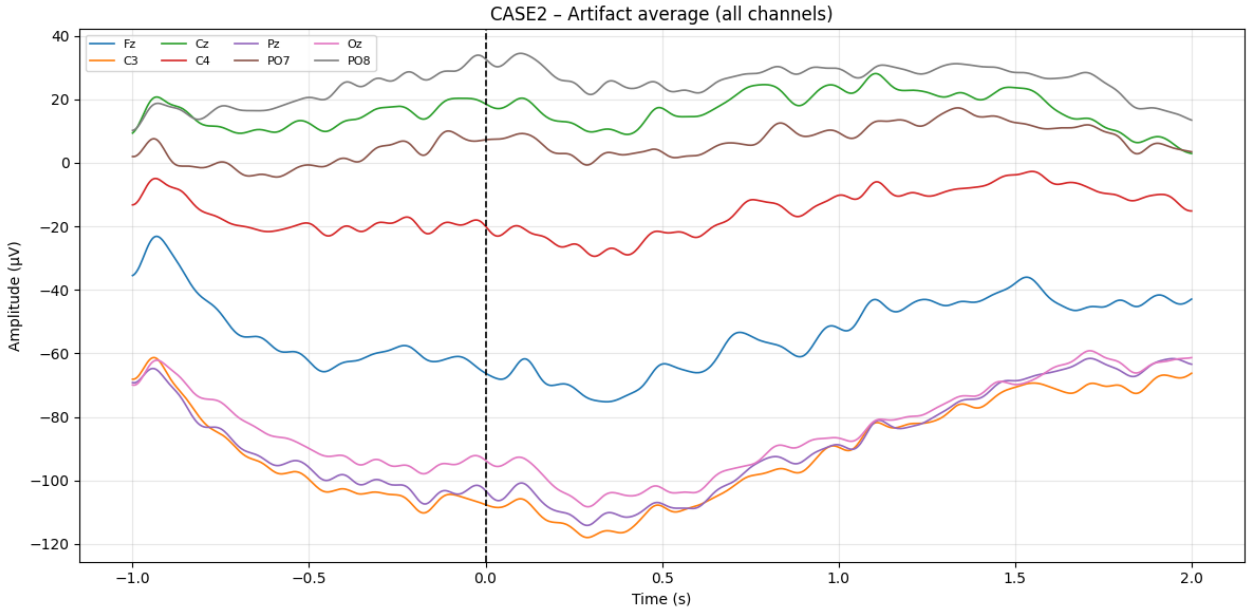
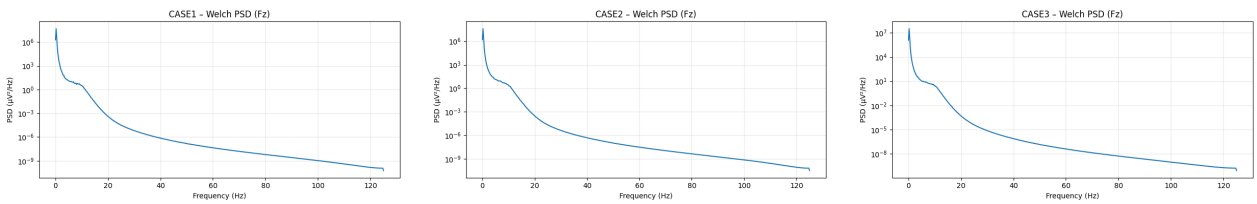

CASE2 – Artifact average (all channels)
Fz
C3
Cz
C4
Pz
PO7
Oz
PO8
Amplitude (µV)
Time (s)
40
20
0
−20
−40
−60
−80
−100
−120
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0

Figure 11

4.6 Power Spectral Density

A Welch PSD estimate was computed for the Fz channel of each recording (segment length 1024 samples, 50% overlap).



(a) Case 1 (b) Case 2 (c) Case 3

Figure 12

5 Summary

Case	Duration (min)	RP events	RP clean	CNV pairs	CNV clean
Case 1	8.08	3	2	0	0
Case 2	11.15	0	0	0	0
Case 3	8.38	7	3	2	1

The limited number of clean epochs (2 for RP in Case 1, 3 for RP and 1 for CNV in Case 3) highlights the need for more annotated trials per participant in future recordings to obtain stable, interpretable ERP averages.